From Γ and φ to New Golden Mathematics

A Plain-Language Map of AMLD, Predator, Settlement Distance, and the Halo(0) Experiment

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Abstract

This document gives a line-by-line conceptual picture of the Automated Mathematical Logical Decider (AMLD) program. It begins with an axiom/theorem bank Γ and a statement φ to be settled. Settlement is broader than ordinary theorem proving: the system may prove φ, prove $\neg\varphi$, establish an appropriate independence result, or expose inconsistency in the relevant assumptions. It then introduces the shared BANK, external verification, Predator as the proof hunter, proof-search states $x \in X$, shortest-settlement distance, the Controlled Abel–Bellman Settlement Theorem, the Half-Step Compass Theorem, Checked Self-State Compass Sufficiency, Tegmark-inspired levels of ATP self-awareness, and the current Halo(0) benchmark. The final goal is a system that can propose a formally verified settlement of a conjecture for which humanity has no golden answer in advance, after which human mathematicians scrutinize the result before it becomes new golden mathematics.

1 Start with Γ

Let

$$\Gamma = \{\text{axioms, definitions, and already accepted theorems}\}.$$

Think of Γ as everything the machine is allowed to assume at the beginning. It defines the mathematical world in which the question will be asked.

For an arithmetic problem, Γ might contain arithmetic axioms. For the Halo(0) experiment, it contains the formal material needed to talk about sets, real numbers, infinitesimals, and the relevant hyperreal construction.

2 Give the machine a statement φ

Let

$$\varphi = \text{the statement we want settled.}$$

The central question is not merely "Can you prove φ?" It is:

> **What is the formal status of φ relative to Γ?**

That broader question is what **settlement** means.

3 What does it mean to settle φ?

There are four operational destinations in the AMLD architecture.

P – Prover

Find a proof that

$$\Gamma \vdash \varphi.$$

In plain English: starting from the permitted assumptions, prove that φ is a theorem.

R – Refuter

Find a proof that

$$\Gamma \vdash \neg\varphi.$$

In plain English: prove that the negation of φ is a theorem.

I – Prover of Independence

Establish, at the appropriate metamathematical level, that

$$\Gamma \nvdash \varphi \qquad \text{and} \qquad \Gamma \nvdash \neg\varphi.$$

In plain English: the axioms do not decide the question.

A clean toy example is a propositional symbol p relative to the empty set of nonlogical axioms:

$$\varnothing \nvdash p, \qquad \varnothing \nvdash \neg p.$$

The intended lesson is that a statement can be independent of a chosen assumption set. Using an expression such as $1 + 1 = 2$ with literally no axioms is less clean, because without background definitions and axioms the symbols do not yet carry their usual arithmetic meaning.

C – Prover of Contradiction / Inconsistency

The cleanest distinct role for C is to search for inconsistency in Γ, in the current BANK, or in some explicitly designated package of assumptions. A contradiction has the form

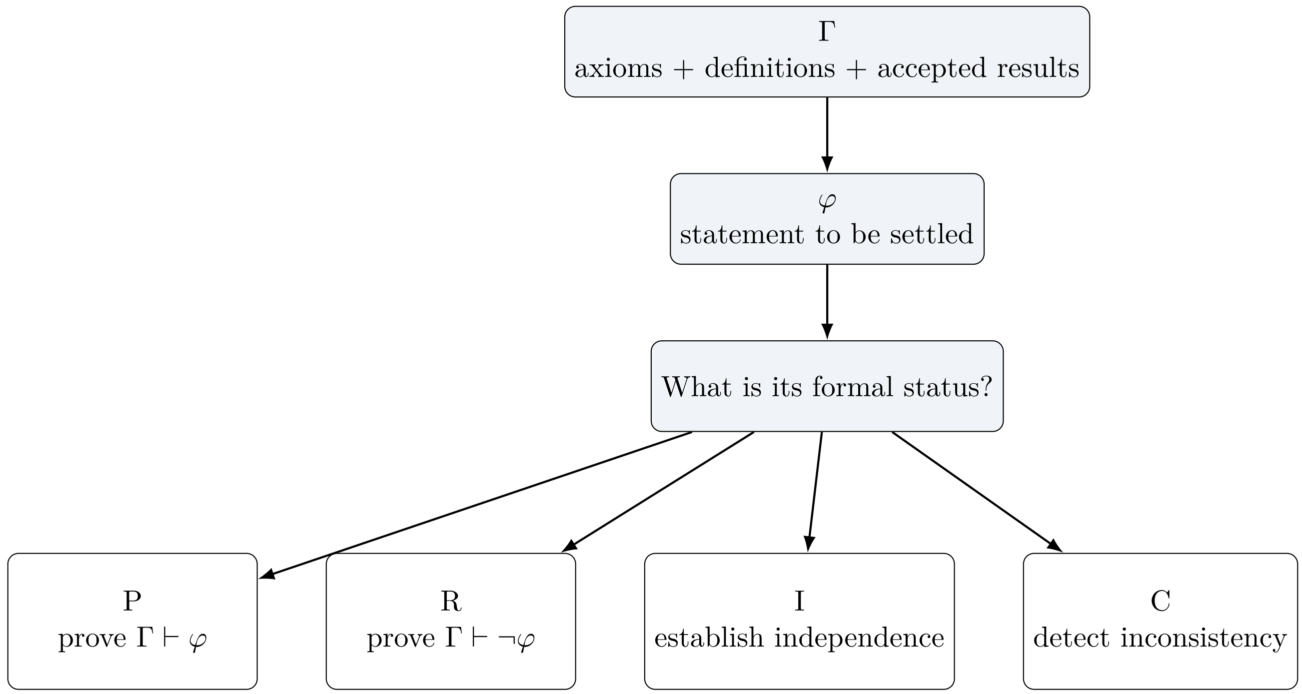
$$\Gamma' \vdash \bot.$$

There is an important logical subtlety: if one instead defines C merely as proving

$$\Gamma \cup \{\varphi\} \vdash \bot,$$

then in ordinary classical logic this often collapses into the refuter's job, because it generally yields $\Gamma \vdash \neg\varphi$. Treating C as an integrity watchdog keeps the four-agent architecture conceptually distinct.

Diagram 1: Settlement branches



4 The four agents share a BANK

Instead of one program trying one thing, imagine four specialists:

$$P,\ R,\ I,\ C.$$

They have different final objectives, but they are not completely isolated. They share a **BANK** of verified information.

Initially,

$$B_0 = \Gamma.$$

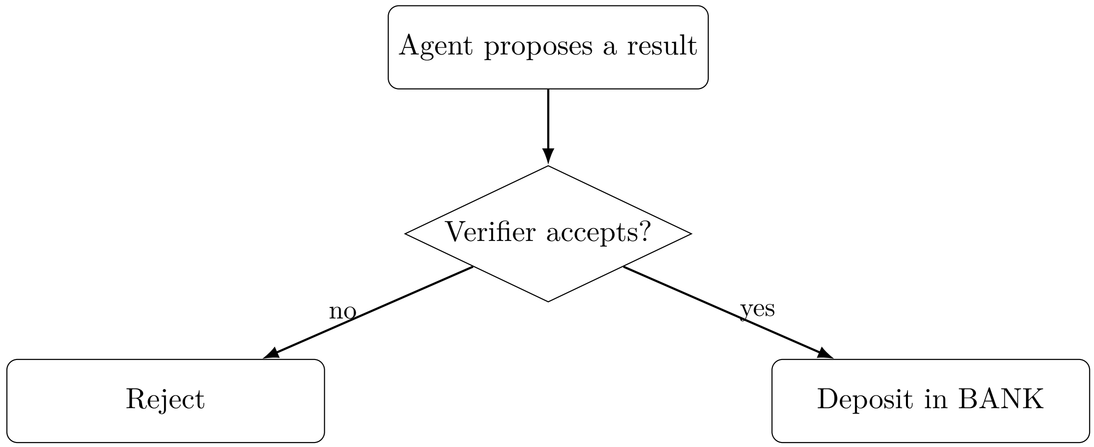
As search proceeds, agents may discover useful intermediate results. A result may not settle φ, but if it has been checked, it can be deposited into the BANK. Schematically,

$$B_0 \subseteq B_1 \subseteq B_2 \subseteq \cdots .$$

This creates an important possibility: **P can discover a lemma that eventually helps R win**. Likewise, R may discover something useful to P, or either may expose structure that eventually helps I. The agents compete over the final settlement but can cooperate through verified information.

5 Verification comes before trust

An agent does not get to declare a theorem merely because it found something that looks convincing. A proposed result must pass a checker before it becomes trusted BANK content.



The design principle is:

Search may be imaginative. Verification must be unforgiving.

Creativity can propose. Verification decides what becomes mathematics.

6 What Predator does

Predator is our automated theorem prover: the part that actually hunts for a proof. Given a formally stated target theorem and a collection of permitted axioms, definitions, and previously established theorems, Predator explores possible sequences of valid deductions.

Predator is not the final judge of its own success. If it believes it has completed a proof, it outputs a formal proof certificate. That certificate is handed to an external checker.

In the present Halo(0) experiment, the external checker is **Metamath**. **Metamath is not part of our system.** It is an independently developed formal mathematics system used as an external verifier. This separation is intentional: the proof hunter does not grade its own homework.

7 Proof search as movement through a state space

Let

$$X = \{\text{all admissible proof-search states}\}.$$

The current condition of the search is some

$$x \in X.$$

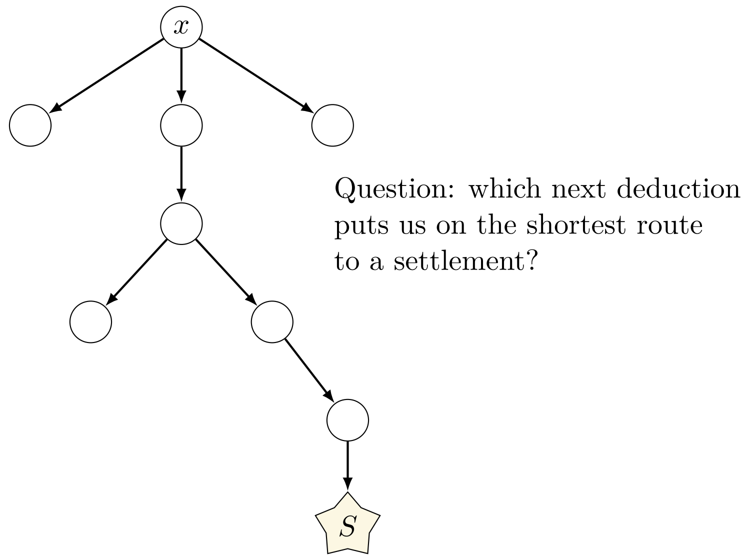
A state may encode open goals, substitutions, partial proof structure, available BANK lemmas, search history, and computational resources already spent.

A legal deduction moves the search to another state:

$$x \longrightarrow x'.$$

A completed verified result corresponds to a special terminal state, which we call a **settlement state** $S \in X$.

Diagram 2: Proof search becomes navigation



8 Shortest settlement and the three main theorems

Ideally, define

$$r^*(x) = \text{minimum remaining cost needed to reach a valid settlement from } x.$$

If every deduction has unit cost, $r^*(x)$ is essentially the length of a shortest remaining settlement route. At a settlement,

$$r^*(S) = 0.$$

The three shortest-settlement theorems form a logical progression.

8.1 Controlled Abel–Bellman Settlement Theorem: define optimal control

Suppose an action a taken at state x produces a successor state $T_a(x)$. If the true remaining settlement value r^* were known, an ideal next move minimizes

$$c(x, a) + r^*(T_a(x)),$$

where $c(x, a)$ is the immediate cost of taking action a.

In plain English:

> **Choose the next deduction whose immediate cost plus remaining distance to settlement is smallest.**

This theorem is foundational because it says what perfect settlement navigation means.

8.2 Half-Step Compass Theorem: when an approximation is enough

In a hard proof search, the exact r^* is generally unavailable. So construct a computable estimate

$$\widehat{r}(x).$$

The Half-Step Compass Theorem says, roughly, that if this estimate is accurate enough – within the relevant half-step margin – then selecting the apparently best successor still selects the genuinely shortest-settlement direction.

The conceptual bridge is

$$\text{ideal Bellman control} \longrightarrow \text{accurate enough surrogate } \widehat{r} \longrightarrow \text{correct next move.}$$

This is why the Half-Step Compass result is the strongest general practical shortest-settlement theorem of the three.

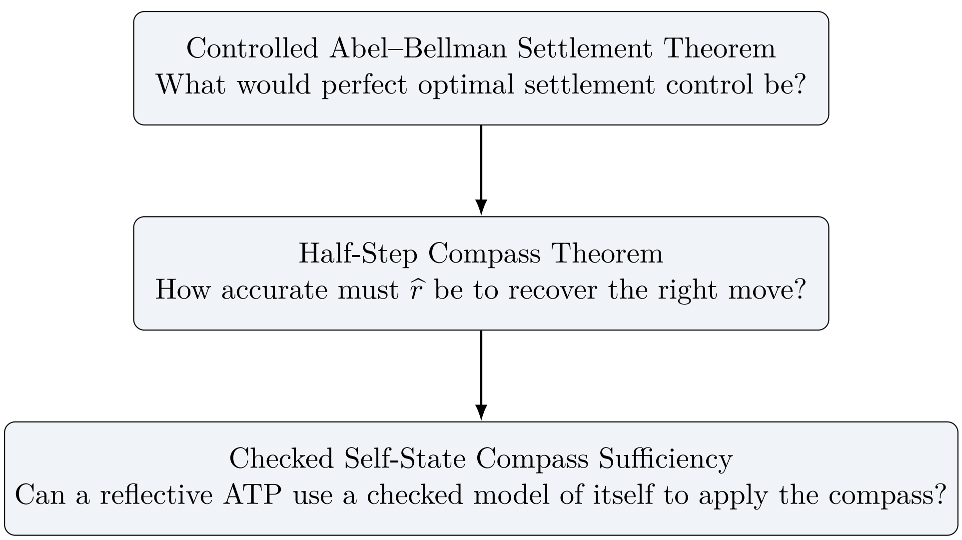
8.3 Checked Self-State Compass Sufficiency: reflective use of the compass

Now suppose the ATP has a formally checked description of its own current search state, say $\mathrm{Self}(x)$. This might encode its open goals, BANK contents, active branches, recent failures, resource use, and estimated settlement values.

The reflective theorem says, in effect:

> If the self-state representation is trustworthy and contains enough information to evaluate the settlement compass, then the ATP can safely use that checked model of itself to choose settlement-directed actions.

So the three results fit together in dependency order as follows.



9 Tegmark-inspired ATP self-awareness

The self-awareness language here is inspired by formal self-modeling ideas associated with Max Tegmark; it is not a claim that the ATP is conscious.

Level 0.
The ATP searches with essentially no usable model of itself: problem $\rightarrow$ search $\rightarrow$ result.

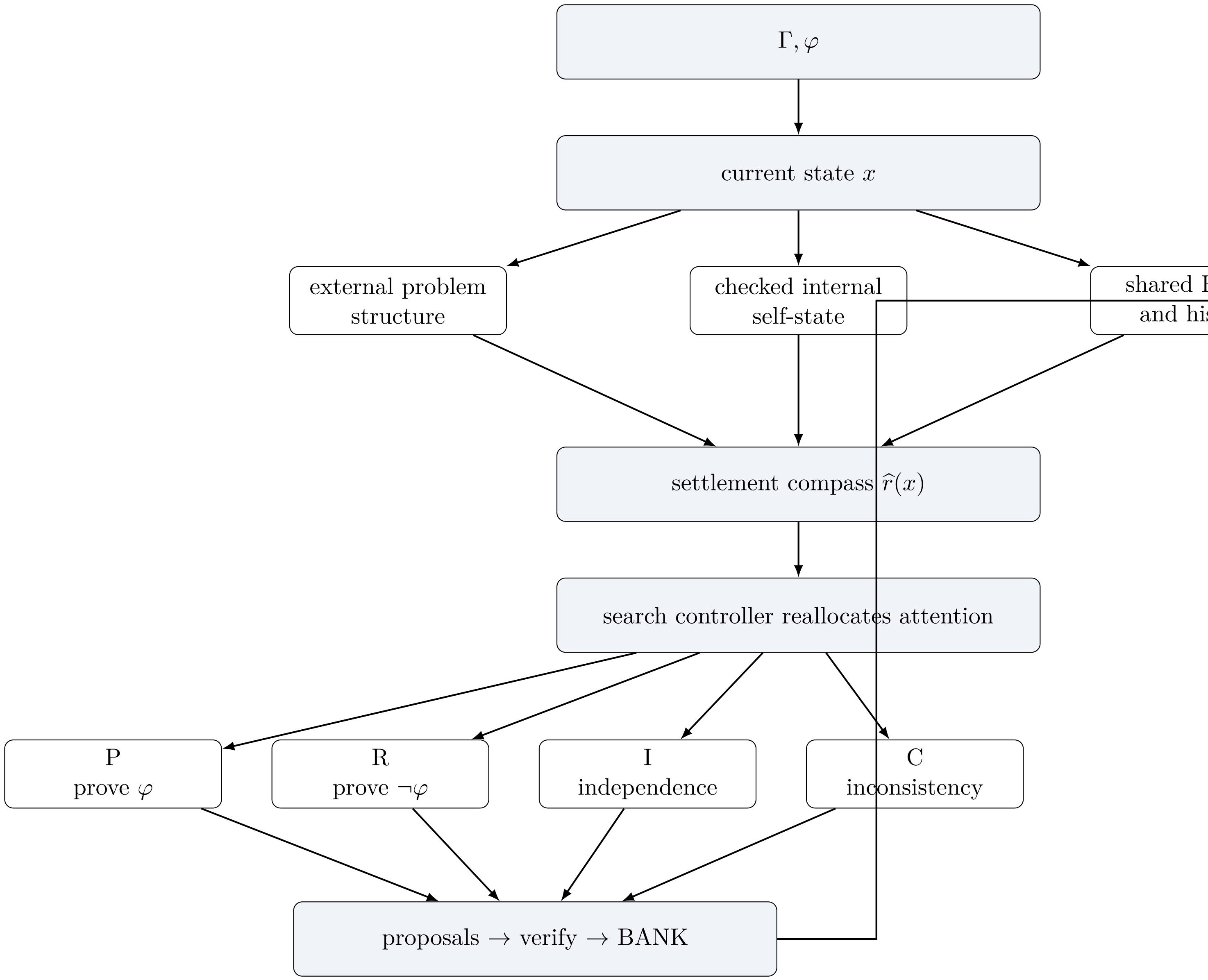
Level 1.
The ATP can observe or record aspects of its own behavior, for example: "I have tried this kind of branch 8,000 times."

Level 2.
The ATP has a usable model of its own search process and uses that model to alter what it does. For example: "P is making little settlement progress; R has recently produced useful BANK lemmas; my settlement estimates favor reallocating effort toward R."

Level ω.
Indefinitely iterated formal reflection or self-modeling. This is theoretically interesting but may be unnecessary or counterproductive for practical proof search.

The working practical hypothesis is that **Level 2 is a useful sweet spot**: enough reflection to improve search control, without requiring endless reflection about reflection.

Diagram 3: Level-2-style AMLD feedback loop



The feedback loop is

$$\text{search} \to \text{observe} \to \text{estimate distance} \to \text{change search} \to \text{search again.}$$

10 Where Halo(0) fits

The current benchmark target is approximately

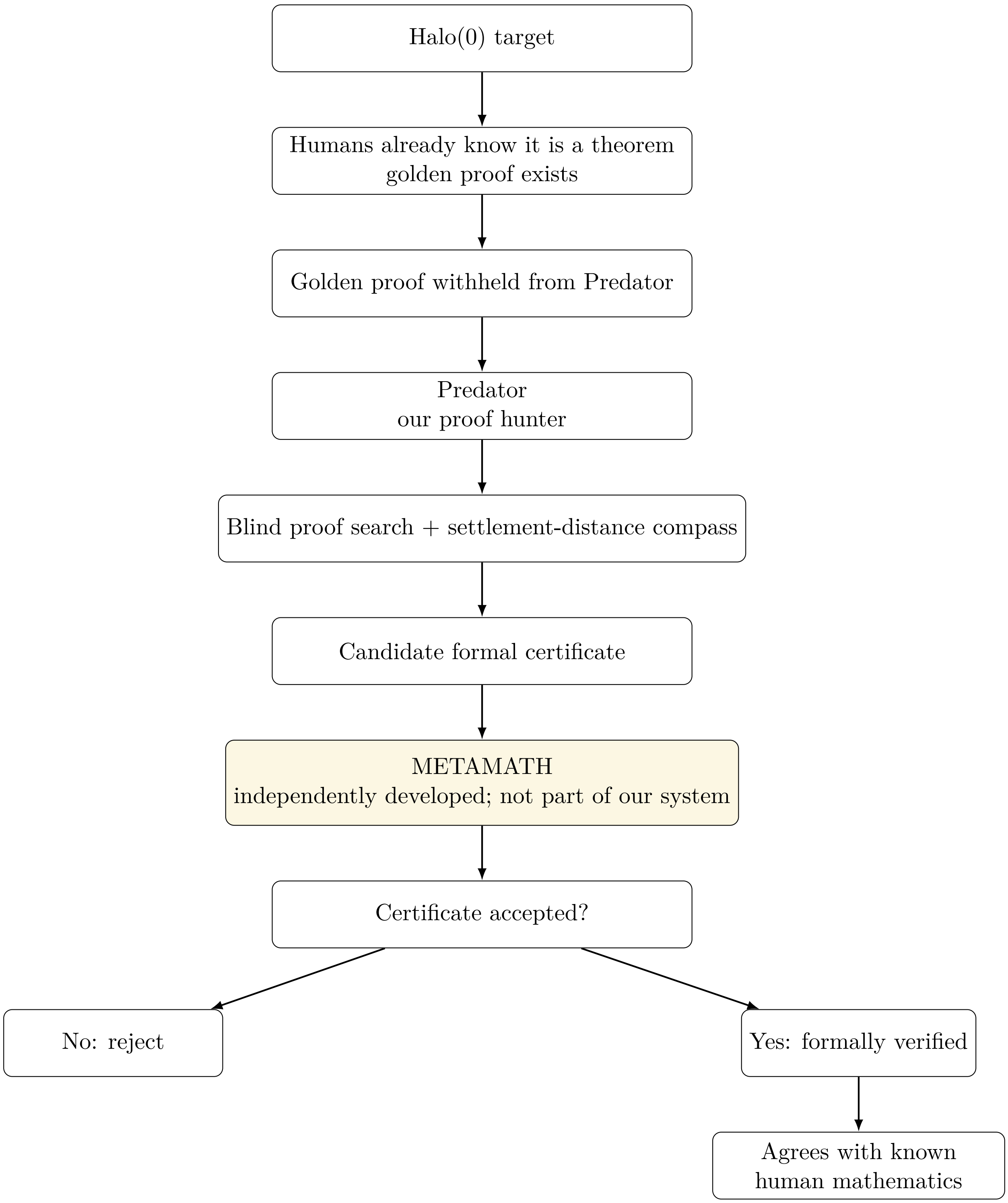
$$\mathrm{hal}(0) \sim \mathbb{R},$$

meaning that the infinitesimal halo around zero has the same cardinality as the real line. In other words, there is a bijection between the infinitesimals in the halo and the ordinary real numbers.

Humans already know this result is true. A **golden human proof** exists. Therefore, in this benchmark we already know that the correct settlement category is the P branch: theoremhood.

The golden proof is deliberately withheld from Predator. The question is not whether the theorem is true; humanity already knows that. The experimental question is whether Predator can independently rediscover a formal proof and emit a certificate that an external verifier accepts.

Diagram 4: Present Halo(0) benchmark



11 Where the Halo(0) experiment currently stands

The experiment has passed several important milestones.

1. The formal Halo target has been constructed.
2. The grammar and verification machinery works.
3. Early infrastructure errors were identified as infrastructure errors rather than being misreported as mathematical failures.
4. Genuine blind proof-search runs have executed successfully as computations.
5. A valid earlier search reached its time limit without emitting a certificate. That outcome is **UNKNOWN**, not evidence that the theorem is false.
6. The current search uses **Predator 8.002C**, whose primary navigation signal is an explicit settlement-distance surrogate $\widehat{r}(x)$.
7. The current implementation does **not** claim that its specific $\widehat{r}$ already satisfies the Half-Step Compass hypothesis. The theorem motivates the architecture, while the Halo runs empirically test the practical surrogate.
8. Hourly Halo search continuation has been resumed under verifier-first stopping rules.

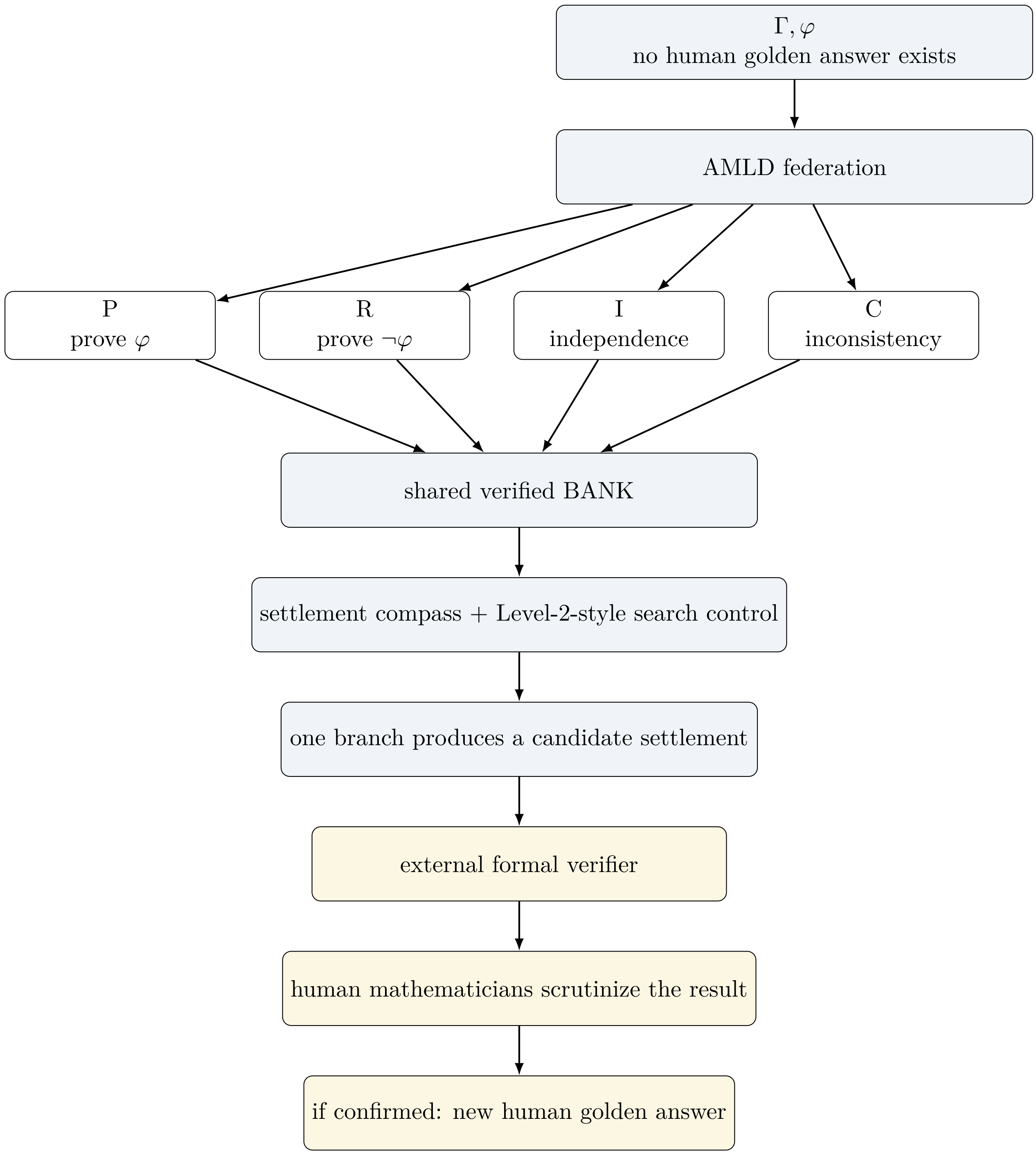
Related controlled experiments have also successfully exercised pieces of the larger architecture, including creativity/meta-compass mechanisms and quotient/value representations. Those successful test runs do not prove Halo(0); they test machinery that may ultimately be useful in harder settlements.

12 The ultimate experiment: no golden answer in advance

Halo(0) is intentionally a controlled benchmark because humanity already knows which branch should win. The ultimate AMLD experiment removes that safety net.

Start with a statement φ for which humans genuinely do not know whether P, R, I, or C will settle the problem.

Diagram 5: Unknown conjecture



Even if an external formal verifier accepts the machine's certificate, the scientific process is not finished for a genuinely open conjecture. Human mathematicians should independently examine the result. Their eventual conceptual proof may be much shorter and may use very different intermediate steps. The paths do not have to match; the settlement must match.

13 The entire research idea in one chain

The program can be compressed into the following sequence:

1. Begin with $\boxed{\Gamma, \varphi}$.
2. Ask for **settlement**, not merely proof.
3. Let P, R, I, C pursue the possible formal outcomes.
4. Let the agents share only verified discoveries through the BANK.
5. Represent the evolving search as a state $x \in X$.
6. Define the ideal shortest remaining settlement value $r^*(x)$.
7. Use Abel–Bellman reasoning to characterize optimal settlement control.
8. Replace unknowable r^* by a practical compass $\widehat{r}(x)$.
9. Use the Half-Step Compass Theorem to understand when approximation is sufficient.
10. Give a reflective ATP a checked representation of its own state.
11. Use Checked Self-State Compass Sufficiency to connect self-modeling with settlement navigation.
12. Aim for useful Tegmark-inspired Level-2-style reflection rather than unbounded reflection.
13. Let Predator perform the actual proof hunt.
14. Let an external verifier – not Predator – check any certificate.
15. First test the architecture against hidden human golden proofs such as Halo(0).
16. Eventually remove the golden answer entirely.
17. Let the machine propose a previously unknown settlement.
18. Require independent formal verification.
19. Require human mathematical scrutiny.
20. If confirmed, the result becomes $\boxed{\text{new golden mathematics}}$.

14 One-sentence summary

The project treats automated mathematics as a verifier-gated search for the shortest route from a formal starting state to one of several legitimate settlement states, first on problems with hidden human golden answers and eventually on conjectures for which no golden answer yet exists.